

# EOD TASK COMPLETION REPORT

**Date:** 31-Aug-2026

**Project:** AWS VPC Networking and EC2

**Status:** Completed

## 1. Create VPC with 2 Private and 2 Public Subnets

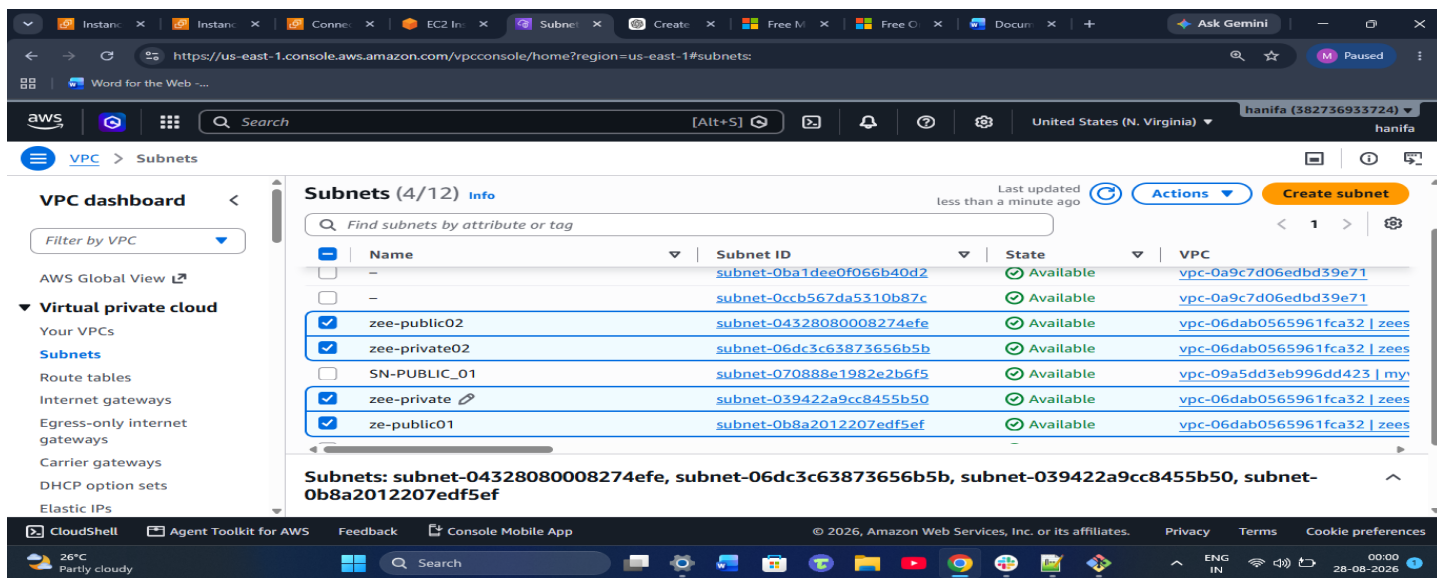
Created an AWS VPC with CIDR:

192.168.0.0/16

Created four subnets:

- Public-Subnet-1 — 192.168.1.0/24
- Public-Subnet-2 — 192.168.2.0/24
- Private-Subnet-1 — 192.168.3.0/24
- Private-Subnet-2 — 192.168.4.0/24

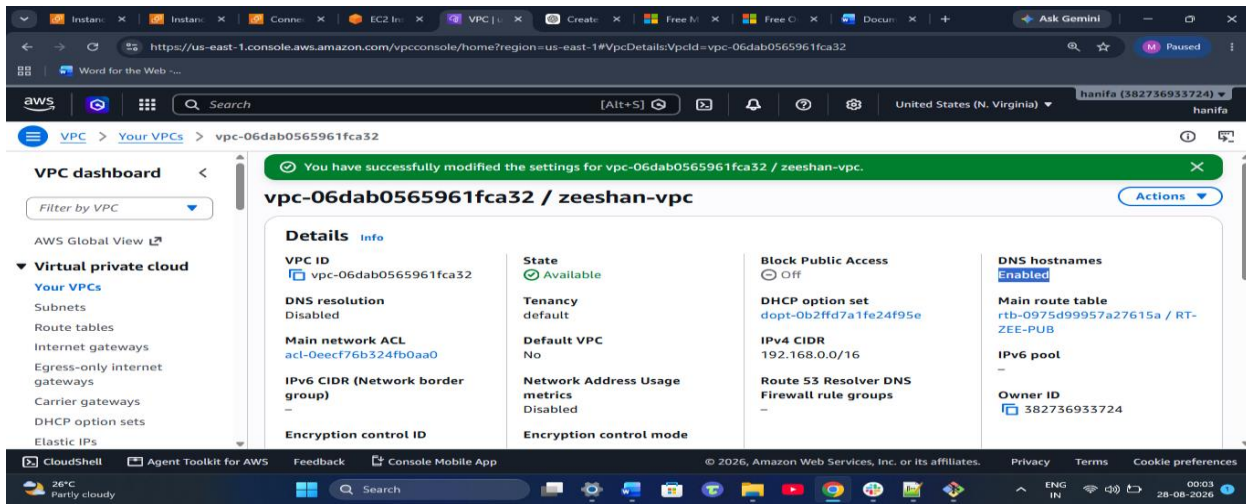
**Evidence:**



## 2. Enable DNS Hostname in VPC

Enabled **DNS Hostnames** in the VPC settings.

**Evidence**

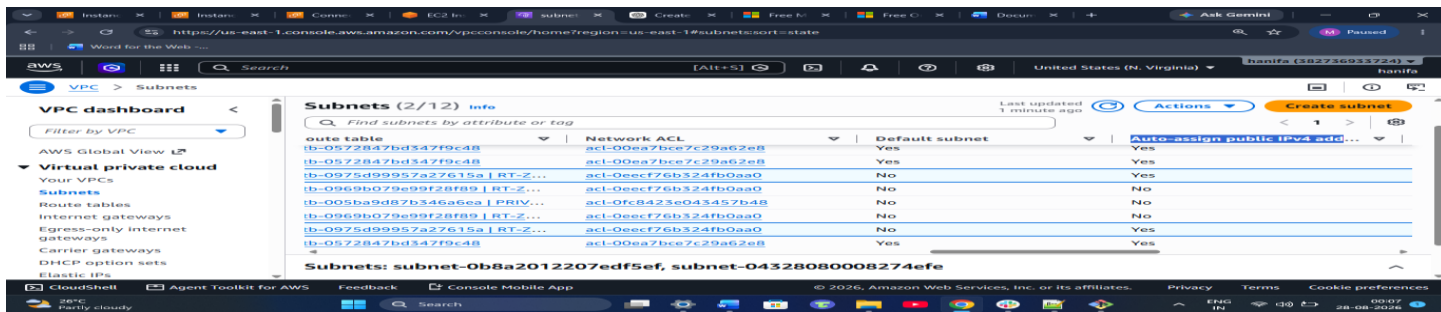


### 3. Enable Auto Assign Public IP in 2 Public Subnets

Enabled Auto-assign Public IPv4 address for:

- Public-Subnet-1
- Public-Subnet-2

Evidence



### 4. Add 2 Private Subnets to Private Route Table

Created a **Private Route Table** and associated:

- Private-Subnet-1
- Private-Subnet-2

|                                     | Name          | Subnet ID                                | IPv4 CIDR      | IPv6 CIDR | Route table ID                                     |
|-------------------------------------|---------------|------------------------------------------|----------------|-----------|----------------------------------------------------|
| <input type="checkbox"/>            | zee-public02  | <a href="#">subnet-04328080008274efe</a> | 192.168.2.0/24 | -         | <a href="#">rtb-0975d99957a27615a / RT-ZEE-PUB</a> |
| <input checked="" type="checkbox"/> | zee-private02 | <a href="#">subnet-06dc3c63873656b5b</a> | 192.168.4.0/24 | -         | <a href="#">rtb-0969b079e99f28f89 / RT-ZEE-PRV</a> |
| <input checked="" type="checkbox"/> | zee-private   | <a href="#">subnet-039422a9cc8455b50</a> | 192.168.3.0/24 | -         | <a href="#">rtb-0969b079e99f28f89 / RT-ZEE-PRV</a> |
| <input type="checkbox"/>            | zee-public01  | <a href="#">subnet-0b8a2012207edf5ef</a> | 192.168.1.0/24 | -         | <a href="#">rtb-0975d99957a27615a / RT-ZEE-PUB</a> |

### 5. Add 2 Public Subnets to Public Route Table

Created a **Public Route Table** and associated:

- Public-Subnet-1
- Public-Subnet-2

| <input checked="" type="checkbox"/> | Name          | Subnet ID                                | IPv4 CIDR      | IPv6 CIDR | Route table ID                                   |
|-------------------------------------|---------------|------------------------------------------|----------------|-----------|--------------------------------------------------|
| <input checked="" type="checkbox"/> | zee-public02  | <a href="#">subnet-04328080008274efe</a> | 192.168.2.0/24 | -         | <a href="#">rtb-0975d99957a27615a / RT-ZEE-</a>  |
| <input type="checkbox"/>            | zee-private02 | <a href="#">subnet-06dc3c63873656b5b</a> | 192.168.4.0/24 | -         | <a href="#">rtb-0969b079e99f28f89 / RT-ZEE-f</a> |
| <input type="checkbox"/>            | zee-private   | <a href="#">subnet-039422a9cc8455b50</a> | 192.168.3.0/24 | -         | <a href="#">rtb-0969b079e99f28f89 / RT-ZEE-f</a> |
| <input checked="" type="checkbox"/> | ze-public01   | <a href="#">subnet-0b8a2012207edf5ef</a> | 192.168.1.0/24 | -         | <a href="#">rtb-0975d99957a27615a / RT-ZEE-</a>  |

## 6. Create EC2 in Public Subnet with t2.micro and Install PHP

Created a t2.micro EC2 instance in the public subnet with a public IP address.

Installed PHP on the EC2 instance.

Example command:

```
sudo yum install php -y
```

Verified the installation:

```
php --version
```

**Evidence:**

```
php8.5-8.5.9-1.amzn2023.0.1.x86_64
php8.5-common-8.5.9-1.amzn2023.0.1.x86_64
php8.5-mbstring-8.5.9-1.amzn2023.0.1.x86_64
php8.5-process-8.5.9-1.amzn2023.0.1.x86_64
php8.5-xml-8.5.9-1.amzn2023.0.1.x86_64
php8.5-cli-8.5.9-1.amzn2023.0.1.x86_64
php8.5-fpm-8.5.9-1.amzn2023.0.1.x86_64
php8.5-pdo-8.5.9-1.amzn2023.0.1.x86_64
php8.5-sodium-8.5.9-1.amzn2023.0.1.x86_64

complete!
root@ip-192-168-1-110 ~]# php --version
HP 8.5.9 (cli) (built: Jul 28 2026 13:06:52) (NTS gcc x86_64)
copyright (c) The PHP Group
uilt by Amazon Linux
end Engine v4.5.9, Copyright (c) Zend Technologies
with Zend OPcache v8.5.9, Copyright (c), by Zend Technologies
root@ip-192-168-1-110 ~]#
```

## 7. Configure NAT Gateway in Public Subnet and Connect to Private Instance

Created a **NAT Gateway** in the public subnet and allocated an Elastic IP.

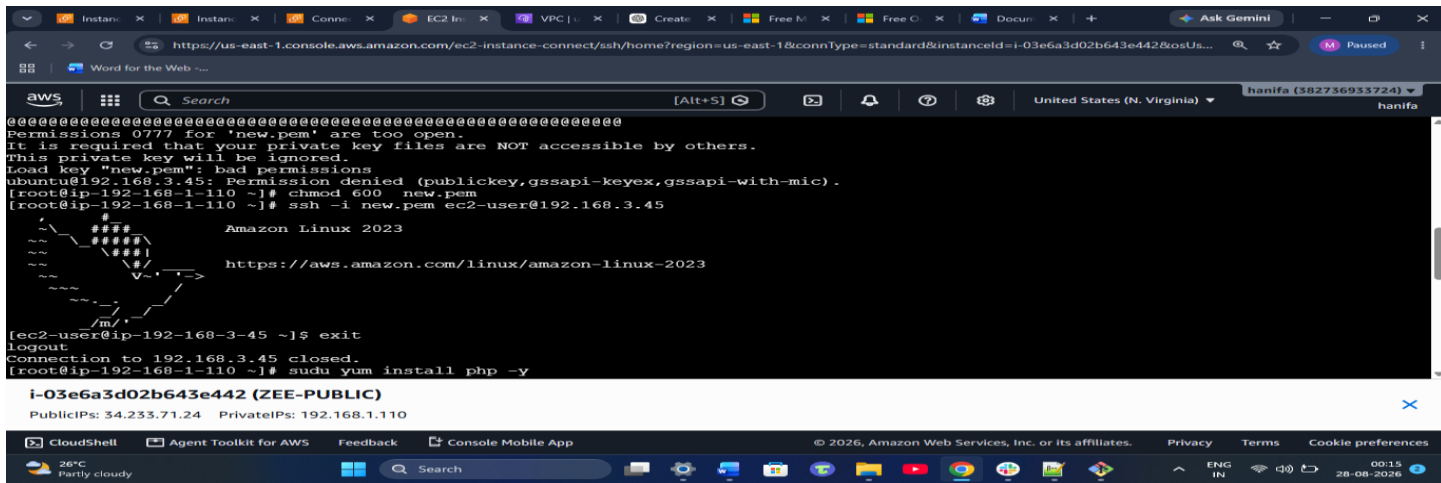
Updated the private route table:

0.0.0.0/0 → NAT Gateway

The private EC2 instance was connected through the public EC2 using SSH:

```
ssh -i new.pem ec2-user@192.168.3.45
```

The private instance uses the NAT Gateway for outbound internet access.



```
aws
Search [Alt+S]
United States (N. Virginia) hanifa (382736933724) hanifa

#####
Permissions 0777 for 'new.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key 'new.pem': bad permissions
ubuntu@ip-192-168-3-45: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[root@ip-192-168-1-110 ~]# chmod 600 new.pem
[root@ip-192-168-1-110 ~]# ssh -i new.pem ec2-user@192.168.3.45

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-192-168-3-45 ~]$ exit
logout
Connection to 192.168.3.45 closed.
[root@ip-192-168-1-110 ~]# sudo yum install php -y

i-03e6a3d02b643e442 (ZEE-PUBLIC)
PublicIPs: 34.233.71.24 PrivateIPs: 192.168.1.110
```

## 8. Install Apache Tomcat in Private EC2 and Deploy Sample App

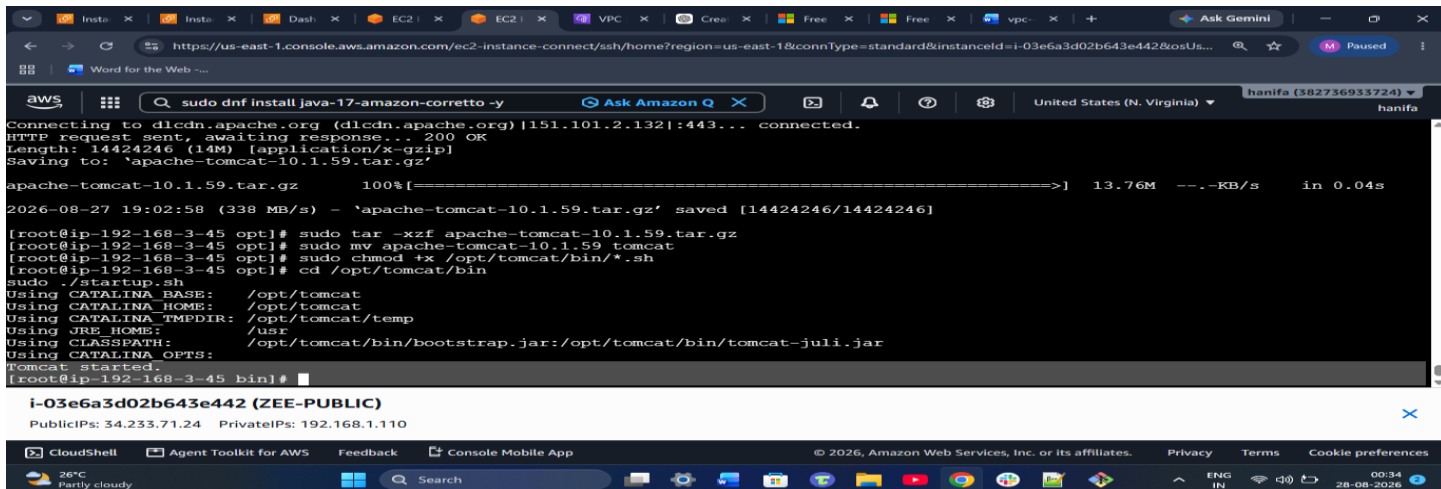
Connected to the private EC2 and installed Java and Apache Tomcat.

Verified Tomcat:

Deployed the sample application under:

/opt/tomcat/webapps/

**Evidence:**



```
aws
Search [Alt+S]
United States (N. Virginia) hanifa (382736933724) hanifa

#####
Connecting to d1c2n.apache.org (d1c2n.apache.org) [151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14424246 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.59.tar.gz'

apache-tomcat-10.1.59.tar.gz  100%[=====] 13.76M  --.-KB/s  in 0.04s

2026-08-27 19:02:58 (338 MB/s) - 'apache-tomcat-10.1.59.tar.gz' saved [14424246/14424246]

[root@ip-192-168-3-45 opt]# sudo tar -xzf apache-tomcat-10.1.59.tar.gz
[root@ip-192-168-3-45 opt]# sudo mv apache-tomcat-10.1.59 tomcat
[root@ip-192-168-3-45 opt]# sudo chmod +x /opt/tomcat/bin/*.sh
[root@ip-192-168-3-45 opt]# cd /opt/tomcat/bin
[root@ip-192-168-3-45 bin]# sudo ./startup.sh
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-192-168-3-45 bin]#
```

## 9. Configure VPC Flow Logs and Store Logs in S3

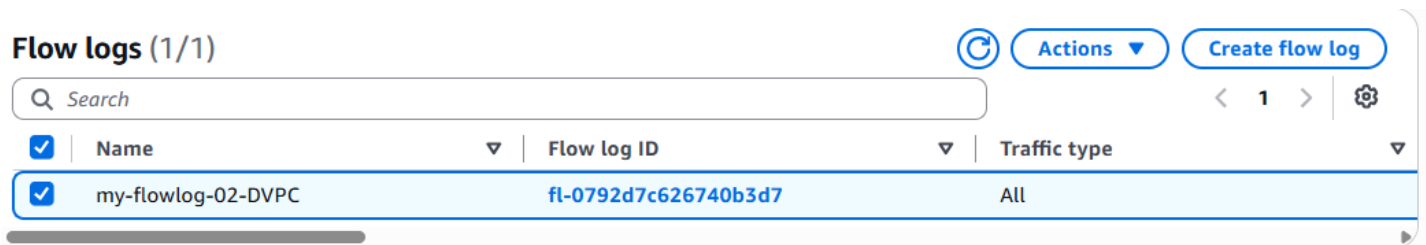
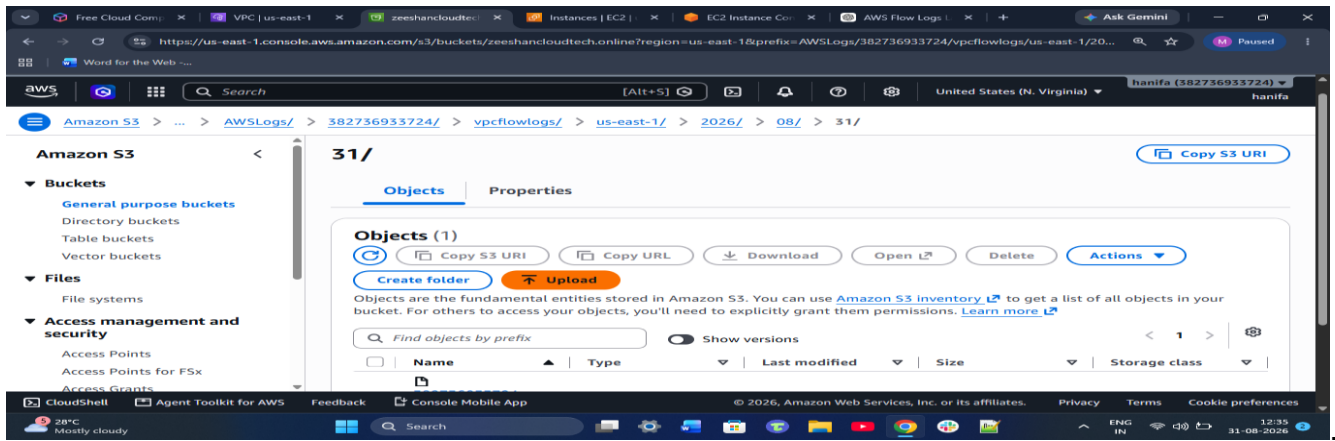
Configured **VPC Flow Logs** with traffic type:

ALL

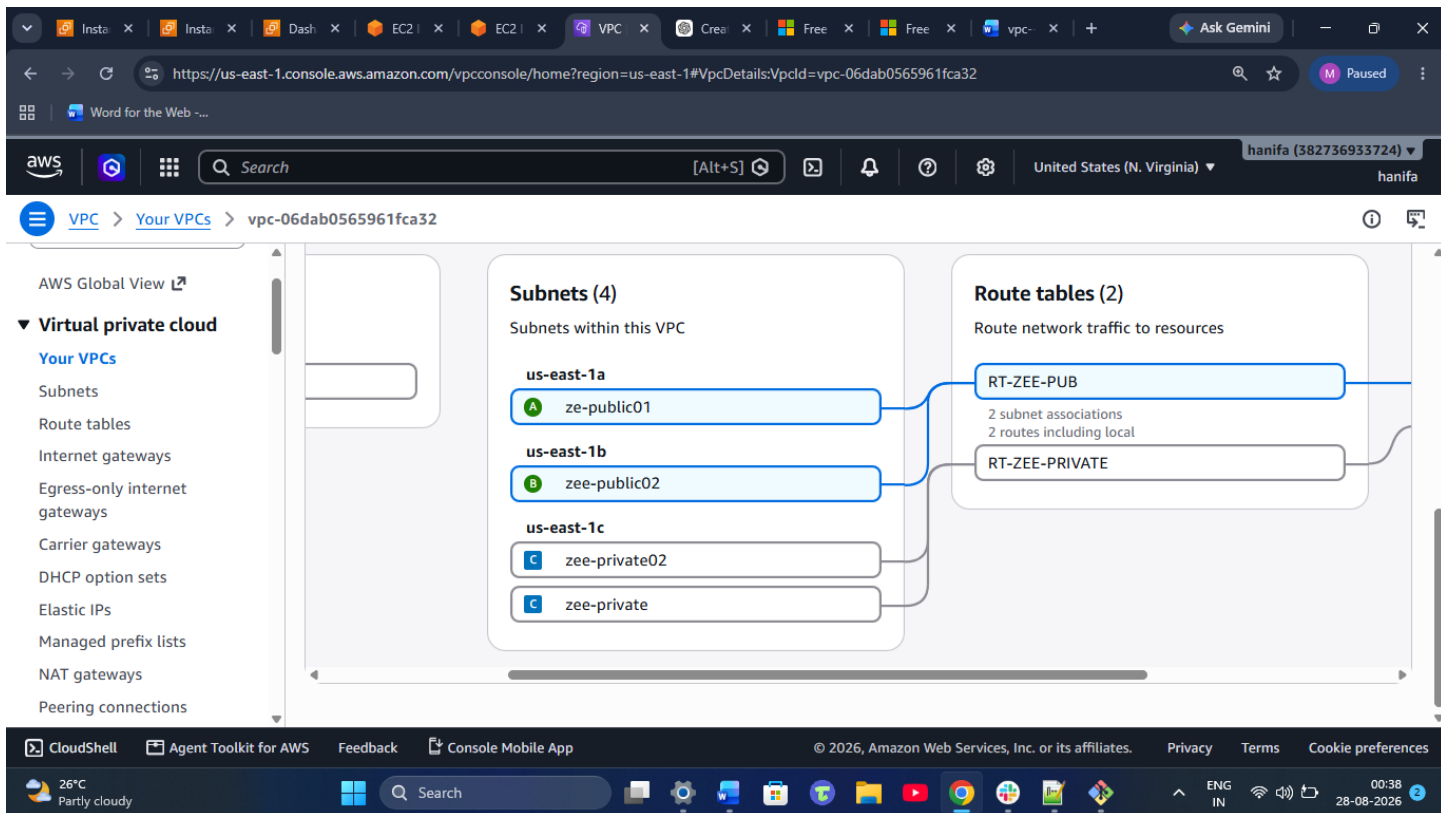
Configured Amazon S3 as the destination.

Verified that VPC Flow Log files were generated in the S3 bucket.

**Evidence**



## 10. Public route table will have the routes to internet and local



Overall Status: Completed